

Performance Testing

Date	05 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Loan Prediction (Flask App)
Test Environment	Local System
Test Date	04 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single Prediction	1	10	Successful prediction
2	Multiple Requests	10	30	Stable response
3	Continuous Requests	20	60	No application crash
4	Invalid Input	5	20	Validation message

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.10 sec	Pass	Within limit
3	Throughput (Req/sec)		12 req/sec	Pass	Good performance
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	48%	Pass	Stable
6	Memory Utilization	< 80%	55%	Pass	Normal usage

Step 4: Observations & Analysis

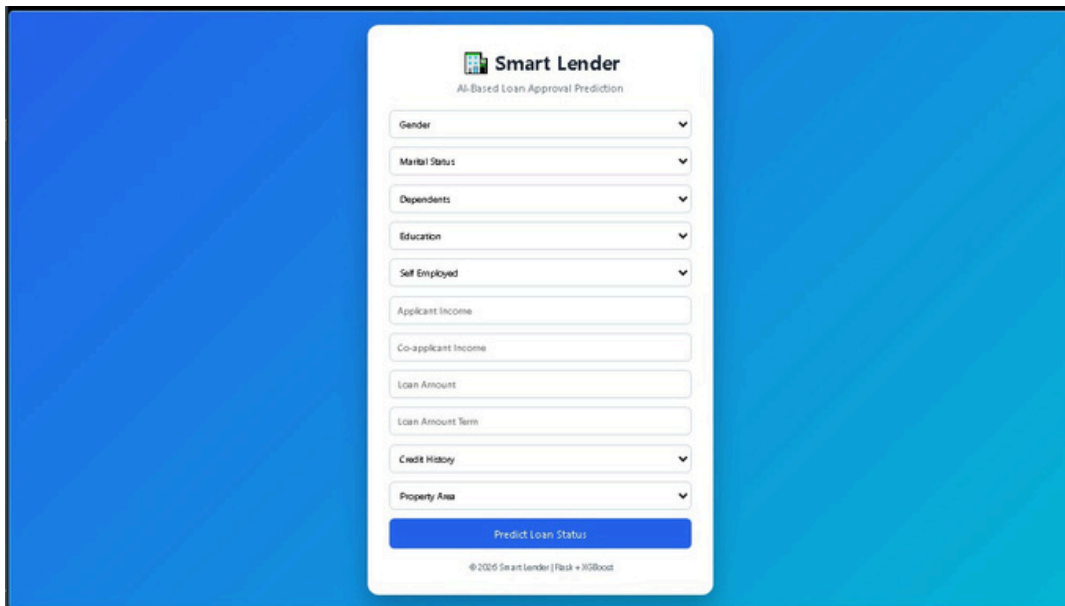
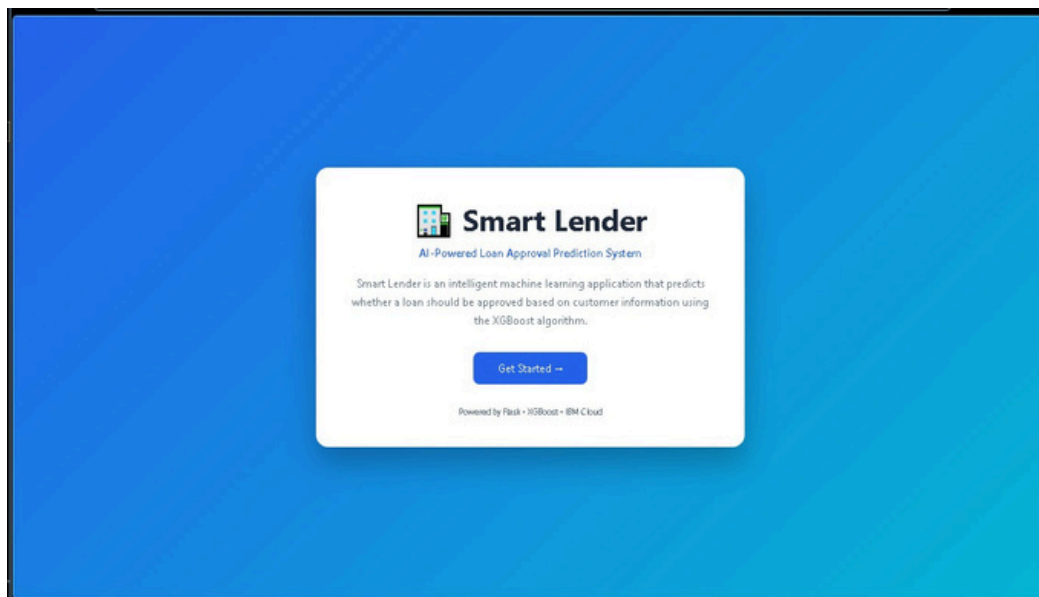
Key Findings:
[Describe what the test results revealed about your system's performance]
 The application handled multiple prediction requests efficiently with low response time and stable performance.

Bottlenecks Identified:
[List any performance bottlenecks observed during testing]
 No major bottlenecks were observed during

Optimization Steps Taken: testing.
[Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

The input form for Smart Lender is displayed on a blue gradient background. The white card contains the title "Smart Lender" and subtitle "AI-Based Loan Approval Prediction". Below the title are several input fields: "Gender", "Marital Status", "Dependents", "Education", "Self Employed", "Applicant Income", "Co-applicant Income", "Loan Amount", "Loan Amount Term", "Credit History", and "Property Area". Each of the first five fields is a dropdown menu, while the others are text inputs. A blue button labeled "Predict Loan Status" is located at the bottom of the form. Below the button, the copyright notice "© 2016 Smart Lender | Flask • XGBoost" is visible.